

From Linux to AI: China's Open Source Revolution: Three Decades of Innovation, Strategy, and Sovereign Technology Leadership

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Abstract

China's open source journey over the past thirty years offers a unique case study in how a government can orchestrate policy, corporate strategy, and developer ecosystems to drive national technology leadership. From early Linux adoption in the 1990s to the rise of cloud computing, AI frameworks, and sovereign software ecosystems today, China has evolved from localized experimentation to global influence. This article examines seven distinct eras of growth, highlighting key milestones, corporate champions, government initiatives, and lessons learned, drawing insights from firsthand experience and engagement with China's open source community.

Disclaimer

The views expressed in this document are solely my own and do not represent those of my current or past employers. If you identify an error or notice a missing critical milestone, please contact me, and I will make the necessary updates.

China's Open Source Journey: A Personal Perspective

China's open source evolution over the past three decades is remarkable in terms of scale, strategic vision, industrial adoption, and technological impact. I have had the privilege of witnessing and contributing to this transformation firsthand. In the early 2000s, while working at Ericsson Research and then the Open Source Development Labs, I traveled extensively to China to promote the adoption of Linux. I collaborated directly with telecom operators, network equipment providers, and ICT companies to enhance Linux for real-world deployments. From 2010 to 2013, during my tenure with the Linux Foundation, I collaborated with Chinese tech companies and communities as they expanded the adoption of open source technologies within their organizations. I delivered dozens of workshops on topics such as using and contributing to open source, license compliance, establishing and managing open source program offices, and leading software transformations. Later, from 2018 to 2025, I worked closely with major Chinese players in the emerging field of open source AI (including Tencent, Baidu, and Huawei), supporting initiatives around AI frameworks, generative models, trusted and responsible AI, and open collaboration ecosystems. Throughout these periods, I interacted with dozens of companies, including China's top tech giants. I witnessed the country's OSS landscape evolve from localized Linux experimentation to a sovereign AI and cloud ecosystem. I remain actively engaged with the Chinese open source community through conference presentations and online webinars that attract audiences of over 300,000 participants. From my vantage point, China's potential for global open source leadership is immense, and its journey, from early Linux adoption to pioneering AI and cloud-native ecosystems, is nothing short of impressive and worth documenting and studying.

This report organizes that journey into seven eras, capturing the milestones of the ecosystem and the lessons learned along the way.

1. **1990–2000: The Linux Awakening:** Early experimentation and the introduction of Linux in academia and research.
2. **2000–2005: Localizing Open Source:** Early adoption, localization, and the emergence of government-backed Linux distributions.
3. **2005–2010: OSS Goes Industrial:** Integration of open source into corporate infrastructure and industrial strategy.
4. **2010–2015: Cloud & Big Data Takeoff:** Adoption of open source cloud and big data technologies, as well as the first significant upstream contributions.
5. **2015–2020: Acceleration & AI Emergence:** Emergence of open source AI frameworks, scaling open source (use and contributions), and strategic national framing of open source criticality as part of national technology priorities.

6. **2020–2025: National Strategy & Sovereignty:** Formalized open source strategies, sovereign platforms, and the growth of OSPOs.
7. **2025 and Beyond: Sovereign & Global OSS** - Looking ahead, we identify a dual approach of domestic ecosystem control and global collaboration.

This structured timeline reflects the evolution of China’s open source ecosystem and my direct observations and contributions during these critical phases, offering insight into how strategic policy, corporate leadership, and grassroots communities combined to make China a global open source powerhouse.

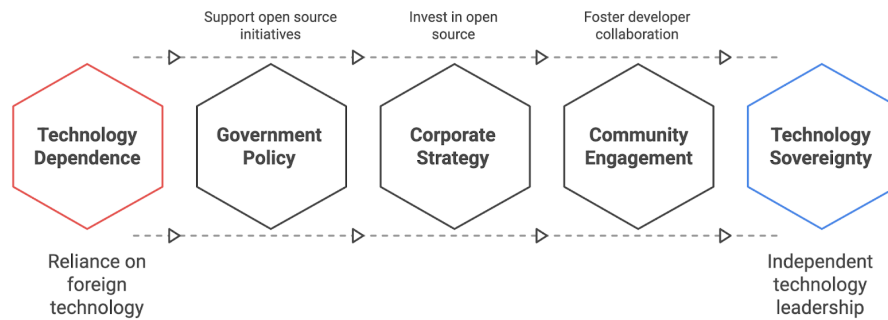


Figure 1: China's open source journey from technology dependence to independent technology

As a side note and before moving to discuss these eras, I would like to highlight recent related posts that I have shared on LinkedIn on the China topic:

- Navigating Open Source Amid Geopolitical Complexity | China OSPO Edition
- China released its 13-point Action Plan on Global AI Governance
- Two AI Action Plans (US and China). What’s the OSPO’s Role?

1990–2000: Introduction and Experimentation

This era marked the introduction of Linux and open source software to China, laying the foundation for experimentation and knowledge transfer primarily in academia.

1997

- The Society for Study, Application, and Development of Free Software was established under the China Software Industry Association in Beijing to promote FOSS.

1999

- COSIX Linux was released by CS&S (China National Computer Software & Technology Service Corporation), a Chinese UNIX/Linux distro, based on Tru64 UNIX.
- TurboLinux (Pacific HiTech) partners with Tsinghua University to produce a Chinese-language version of TurboLinux, targeting government ministries to promote Linux adoption.
- Red Flag Linux was developed at the Institute of Software, Chinese Academy of Sciences, with backing from the government-owned Shanghai NewMargin VC.
- XteamLinux Chinese version 1.0 was released, selling 100,000 copies within months; followed by XteamLinux 3.0 in March 2000 (40,000 copies in two weeks), plus server variants.

2000-2005: Early Adoption and Localization

During this era, China focused on adapting OSS to local needs, building domestic distributions, and fostering early developer communities, while government interest in tech sovereignty emerged.

2000

- The Chinese Academy of Sciences (CAS) has launched Red Flag Linux, a domestic Linux distribution aimed at government and state-owned enterprises.
- At LinuxWorld Asia, China's Minister of Industry and Information Technology signals government policy to use Linux in all ministries. This signal represents early official support for Linux as an alternative to Microsoft Windows, emphasizing its benefits in terms of sovereignty and security.
- Intel and Bluepoint establish a Linux Technical Support Center in Shenzhen with Proof-of-Concept, training, and online support capabilities.
- Red Flag Software is spun off from the Chinese Academy of Sciences; Red Flag Linux Desktop 2.0, Server ISV, and embedded solutions are released.

2001

Government policies encourage the adoption of Linux and open source office suites (like WPS and OpenOffice derivatives) to reduce foreign software dependency.

2002

- Universities and research institutes are beginning to standardize Linux in their academic and research labs.
- The LAMP stack (Linux, Apache, MySQL, PHP/Perl) is growing in popularity among developer communities.

- Small OSS meetups and forums emerge in Beijing and Shanghai.

2003

- Huawei begins serious experimentation with embedded Linux for networking gear, laying the foundation for long-term OSS adoption.
- The government continues boosting Red Flag Linux in ministries.

2004

- Wuhan Deepin (initially Hiweed Linux, later Deepin OS) emerges, targeting Chinese desktop users with localized Linux distributions.
- Developer communities around Java, PHP, and MySQL are maturing.

2005

- OSS enters the early mainstream phase in China.
- Government ministries pilot Red Flag Linux deployments.
- Universities include Linux courses in CS curricula.
- Broader state-led discourse on “information sovereignty” reinforces support for OSS.

2005-2010: Industrial Strategy Phase

During this period, OSS became part of industrial strategy, with large Chinese companies adopting and scaling open source internally while government policies encourage domestic tech competitiveness.

2005

- Open source is framed as a way to strengthen technology sovereignty.
- State agencies begin encouraging open source alternatives in software procurement.
- Grassroots developer groups around Linux, Java, and PHP gain momentum.

2006

- Huawei and ZTE are increasing their use of Linux in networking equipment and telecom infrastructure.
- This period marks the start of serious corporate adoption beyond experiments.

2007

- Growth of community-driven Linux user groups across Beijing, Shanghai, and Shenzhen.

- Universities offer courses on Linux and open source development.

2008

- The Beijing Olympics serve as a catalyst for scaling internet services powered by open-source software.
- Baidu, Alibaba, and Tencent (BAT) push large-scale deployments of Linux, MySQL, and early Hadoop clusters.

2009

- Alibaba launches Aliyun (Alibaba Cloud), which is heavily reliant on open source to build scalable infrastructure. This milestone marks the entry of open source into Chinese cloud computing.

2010

- CSIA frames open source as an *industrial enabler* rather than just a free alternative.
- Tencent integrates open source into its gaming and messaging platforms.
- Apache Hadoop adoption accelerates, with Baidu and Alibaba investing in big data search and recommendation systems.

2010-2015: Cloud, Big Data, and First Serious Upstream Contributions

This era marks China's first profound upstream contributions to global OSS, with cloud, big data, and mobile ecosystems driving both domestic and international engagement

Major Companies

- Alibaba Cloud was built on open source stacks (Hadoop, Spark, MySQL, Kubernetes). Alibaba released JStorm and RocketMQ as open source.
- Tencent created TARS (later open sourced it and transitioned it to the Linux Foundation under the TARS Foundation). Tencent also launched several infrastructure-related projects tied to its WeChat infrastructure.
- Huawei deepened its contributions to the Linux kernel and networking projects used in its products. Additionally, they scaled their contributions to OpenStack in line with its adoption, supporting Huawei Cloud.
- Xiaomi leveraged Android OSS to create MIUI.
- Baidu invested in AI frameworks; early development of PaddlePaddle (2013).

Ecosystem Growth

- GitHub adoption expanded.

- Chinese developers are more visible in global OSS.
- Rise of Deepin Linux as a polished desktop OS alternative.
- The Apache Hadoop ecosystem has been localized and scaled in China. OSS meetups and conferences experienced significant growth, particularly in Beijing, Shanghai, and Shenzhen.

Government / Industry

- MIIT encouraged cloud adoption with OSS underpinnings.
- China Software Industry Association emphasized OSS as part of national software competitiveness.
- The first local policies promoting “ ” (self-controllable) software included OSS in their scope.

Community & Education

- OSS communities around Hadoop, Spark, and Android proliferated.
- University programs (notably Tsinghua, Beihang) encouraged OSS participation.
- Rise of Chinese-language OSS forums and GitHub mirrors.

Global Positioning

- Huawei’s involvement in OpenStack positioned it as one of the first Chinese players in global OSS governance.
- Tencent and Alibaba began engaging with global OSS foundations (Apache Foundation, Linux Foundation) in exploratory ways.
- China positioned open source as both an enabler for the domestic industry and a way to gain international credibility.

2015-2020: Acceleration & AI Emergence

Open source becomes central to AI, cloud, and industrial strategy, with corporate champions and government policy aligning OSS with national competitiveness and sovereignty.

2015

Kubernetes adoption surges across BAT (Baidu, Alibaba, Tencent) and startups. This marks the beginning of China’s cloud-native wave, with Kubernetes serving as the backbone for domestic cloud and AI services.

2016

Baidu open sources PaddlePaddle, marking the first large-scale Chinese deep learning framework to go open source. Baidu then transitioned Elastic Deep

Learning project into the LF AI & Data Foundation, under neutral technical governance, and opened participation in the project’s development.

2017:

- Beijing explicitly ties open source to self-reliance and tech sovereignty: the Cybersecurity Law, the State Council AI Development Plan, and the AI development plan highlight open source ecosystems.
- Alibaba launches Alink (big data ML library). Built for machine learning at scale; reflects emphasis on domestic big data/AI solutions.
- Tencent launched Angel (a distributed ML system)

2018

- Gitee gains traction as a domestic alternative to GitHub and becomes the government-preferred OSS hosting platform.
- Tencent hosts Angel in the LF AI & Data Foundation as part of their strategy to rival Western ML frameworks and signal (along with Baidu) their participation in global open source AI governance while promoting homegrown tech.
- Ant Group releases open source blockchain frameworks. Early push to create indigenous blockchain ecosystems tied to fintech leadership.

2019

- Huawei launches openEuler and openGauss, explicitly positioning these projects as sovereign alternatives to Western software, aligned with Made in China 2025.
- Tencent and Alibaba scale their Kubernetes contributions, given their dependency on the project for their cloud infrastructure. Both companies are major drivers of container and cloud-native innovation across Asia, signaling their capability to co-lead the cloud native computing ecosystem.
- DeepSeek launches an open source AI research framework and quickly becomes one of the most influential Chinese AI labs, utilizing open source to accelerate large-scale AI development.

2020

- The expansion of open data and standards platforms, with state-led initiatives encouraging the development of open standards and data-sharing platforms to support domestic AI and smart city applications.
- COVID-19 accelerates the adoption of open source health and AI solutions, resulting in several domestic projects for epidemiological modeling, AI-based diagnosis, and open datasets gaining prominence.

2020–2025: National Strategy and Sovereignty

China formalizes OSS as a core national strategy, combining domestic sovereignty, AI leadership, and global engagement while promoting controlled open source ecosystems.

Government & Policy

- The US sanctions on Huawei, ZTE, and SMIC accelerate the tech sovereignty push with OSS framed as critical for reducing dependency on foreign technology.
- MIIT formally promotes open source ecosystems (2021).
- CAC implements stricter rules for algorithm transparency and data security, shaping the adoption of OSS.
- OSS is explicitly embedded in national 5-year tech independence plans across AI, semiconductors, and cloud ecosystems.

Foundational Institutions

- OpenAtom Foundation launched in 2020 as China’s central OSS governance hub, hosting projects like OpenHarmony, OpenEuler, AliOS Things, and more.

Major Companies & Projects

- Huawei: HarmonyOS / OpenHarmony launched as open source; OpenEuler and openGauss promoted as sovereign alternatives.
- Baidu: PaddlePaddle flagship AI framework; ERNIE large language models released in controlled open-source format.
- Tencent: Angel ML framework contributed to LF AI & Data; heavy contributions to Kubernetes and cloud-native ecosystems.
- Alibaba: Tongyi Qianwen and Alink for ML and big data, released with open-ish licenses.
- Ant Group: open source blockchain frameworks.
- DeepSeek is positioned as a domestic rival to OpenAI, making significant contributions to the field of generative AI.
- Chinese tech companies formalized Open Source Program Offices (OSPOs) to manage strategy and compliance, aligning corporate strategy with state-backed open source goals.

Ecosystem & AI Advances

- Baidu, Alibaba, Tencent, ByteDance, and DeepSeek integrate open source into large-scale AI models.

- AI frameworks (PaddlePaddle, MindSpore, MegEngine) form the backbone of domestic AI sovereignty.
- Gitee emerges as the government-preferred domestic alternative to GitHub.
- Open source compliance gains attention as laws codify algorithm transparency and licensing requirements.

Global Engagement vs. Decoupling

- Chinese firms are increasing their contributions to international foundations (CNCF, LF AI & Data, Eclipse) while developing parallel domestic ecosystems.
- OpenAtom Foundation is positioned as a Chinese counterpart to US/EU OSS governance, aligning projects like OpenEuler, OpenHarmony, and openGauss
- The balance between global engagement and tech sovereignty drives corporate strategy.

Strategic Trends

- OSS is formally tied to AI, semiconductors, and cloud strategies under the national five-year plans.
- Sovereign OS (OpenEuler, OpenHarmony) and databases (openGauss, OceanBase) are promoted in government and state-owned enterprises.
- AI + LLMs are framed as the next frontier for sovereignty, with DeepSeek exemplifying China’s domestic generative AI ambitions.

2025 and Beyond: Looking Ahead: Sovereign & Global Dual Strategy

OSS is expected to become fully integrated into China’s tech sovereignty strategy, with domestic AI and software ecosystems advancing in parallel to selective global collaboration.

Government & Policy

- Stronger orchestration of open source through OpenAtom Foundation and Gitee, with domestic projects prioritized over foreign platforms like GitHub.
- Policies and licensing frameworks will continue to shape the release of AI models, with an emphasis on transparency, security, and controlled data sharing.

Major Companies & Projects

- Chinese tech giants (Huawei, Alibaba, Tencent, Baidu, ByteDance) will increasingly integrate OSS into sovereign technology stacks.
- Generative AI projects, such as DeepSeek and new domestic models, will proliferate, released under controlled open-source licenses.
- Expansion of sovereign OS, databases, and cloud-native frameworks to support government, state-owned enterprises, and critical infrastructure.

Ecosystem & AI Advances

- China will maintain a dual strategy: actively participating in global foundations (such as the Linux Foundation, Apache, and CNCF) where possible, while creating domestic forks when geopolitical restrictions limit collaboration.
- Cloud-native, AI, and LLM ecosystems will become increasingly domestic-first, with local contributions driving the OSS ecosystem.

Global Engagement vs. Decoupling

- Strategic balancing: domestic sovereignty and self-reliance coexist with selective global engagement for technology learning, influence, and standard-setting.
- China's open source leadership will increasingly be tied to its national independence narrative.

Strategic Trends

- OSS as critical infrastructure: chips, AI, databases, cloud, and software stacks will be framed as foundational for national independence.
- Investments in domestic OSS, AI frameworks, and cloud-native ecosystems are expected to intensify, shaping the next generation of China's tech sovereignty.
- Generative AI and AI model hosting will become key areas where OSS and state strategy intersect.

Lessons Learned

China's multi-decade journey in open source offers a unique perspective on how government policy, corporate strategy, and developer ecosystems can be orchestrated to achieve national technology goals, providing lessons that other countries and organizations can adapt to their own contexts.

1. **Strategic Alignment Between Government and Industry:** China's experience shows that open source adoption is vastly accelerated when government strategy explicitly links OSS to national priorities. By framing open source as a tool for technology sovereignty, AI leadership, and cloud independence, China created incentives for corporations to invest

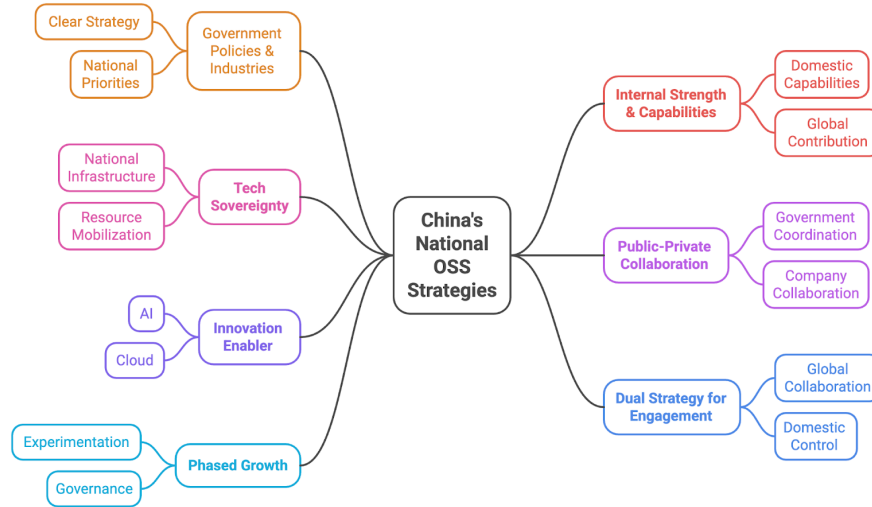


Figure 2: Core elements of China's national open source strategies

in internal OSS projects and infrastructure. This alignment also encouraged universities, research institutes, and private developers to participate in building domestic ecosystems, demonstrating that policy clarity and strategic framing can significantly shape national technology outcomes.

2. **Build Domestic Capabilities Before Contributing Globally:** China followed a phased approach: first, experimentation and adoption of Linux and open source tools; second, internal scaling with cloud-native platforms, databases, and AI frameworks; and only later, selective upstream contributions through foundations such as the Linux Foundation, Apache Foundation, LF AI & Data, and CNCF. This sequence highlights that a country or organization benefits from mastering internal OSS capabilities, creating robust ecosystems, and understanding local needs before prioritizing global contributions.
3. **Sovereignty Drives Investment:** Concerns over reliance on foreign technology drove China to invest heavily in domestic operating systems (OpenEuler, OpenHarmony), databases (openGauss, OceanBase), cloud infrastructure, and AI frameworks (PaddlePaddle, Elastic Deep Learning, MindSpore, Angel). This heavy investment illustrates that framing OSS as part of national digital sovereignty can mobilize resources, attract top talent, and accelerate the development of high-impact technology platforms.
4. **Public-Private Collaboration Multiplies Impact:** China leveraged both private and state-owned enterprises to scale the adoption of open

source rapidly. Companies like Huawei, Alibaba, Tencent, and Baidu became innovation engines for OSS, supported by state guidance and incentives. Public-private collaboration allowed projects to scale nationally while also establishing governance mechanisms (e.g., OpenAtom Foundation). This collaboration underscores that co-investment between government and industry can accelerate adoption, drive standardization, and ensure sustainability.

5. **OSS as a Platform for Innovation:** Open source became the foundation for China’s AI, cloud, and big data ecosystems. OSS enabled the rapid creation of generative AI models (DeepSeek, ERNIE, Tongyi Qianwen), cloud-native infrastructure, and domestic software alternatives. By using OSS as a base rather than building everything from scratch, China accelerated technology cycles and innovation capacity, demonstrating that OSS is not just cost-saving but also a launchpad for next-generation technology leadership.
6. **Balance Global Engagement with Domestic Ecosystem Control:** China shows the value of a dual approach: participating in global OSS communities for learning, collaboration, and influence, while maintaining domestic forks and ecosystems when geopolitical or strategic factors require autonomy. This dual strategy ensures access to global innovations without compromising national sovereignty, illustrating how careful governance can enable countries to benefit from open source technology while protecting their strategic interests.
7. **Phased and Layered Approach Works:** China’s OSS journey illustrates a clear, staged progression: early experimentation → localized adoption → internal scaling → governance structures → global influence. Each stage is built on the previous one, combining capability development, ecosystem building, and governance. This approach highlights the importance of long-term planning and incremental capability development when designing national OSS strategies.

Conclusion

China’s open source evolution demonstrates the power of aligning government policy, corporate strategy, and developer communities to drive innovation and national technology sovereignty. Strategic localization, corporate champions, and active community engagement enabled China to scale from Linux experimentation to leadership in AI and cloud computing. The dual approach, participating globally while maintaining domestic control, offers a model for leveraging open source as a strategic national asset.

About the Author

Dr Ibrahim Haddad is currently Head of Infotainment at Volvo Cars. He is widely recognised for his leadership in building and scaling Open Source Program Offices (OSPOs), having founded and directed them across multiple organisations, and advised both corporations and NGOs on open source strategy and governance. Previously, he served as Vice President of Strategic Programs (AI) at the Linux Foundation, where he was the Executive Director of LF AI & Data and the Founding Executive Director of the PyTorch Foundation. During his last tenure at the Linux Foundation, his core mandate was to support and grow the open source AI ecosystem by providing a neutral, trusted hub for developers and organizations to code, manage, and scale open source AI projects. Before the Linux Foundation, Dr Haddad served as Vice President of R&D and Distinguished Engineer at Samsung Research Silicon Valley. His career includes R&D technical and leadership roles at Ericsson Research, Open Source Development Labs, Motorola, Palm, and Hewlett-Packard. He holds a PhD with honours in Computer Science from Concordia University (Montréal, Canada).

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